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Adaptive multi-mode locomotion for bipedal wheel-legged robots via sparse mixture-of-experts deep reinforcement learning

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The bipedal wheel-legged robot combines the high energy efficiency of wheeled movement with the terrain adaptability of legged locomotion. However, achieving a smooth transition between these two heterogeneous motion modes within a unified control framework remains challenging. This study proposes a reinforcement learning control framework that integrates the Mixture of Experts (MoE) architecture. This approach employs a “divide and conquer” strategy by introducing a dynamic gating network and a Top-K sparse activation mechanism, which automatically allocates different motion modes to specific expert subnetworks, effectively decoupling conflicting gradients. Simulation results demonstrate that, compared to the single-network PPO method, the MoE-enhanced algorithm exhibits significant improvements in training stability and rewards. The learned policy successfully achieved smooth rolling on flat surfaces and transitioned to dynamic leg-lifting gaits when confronted with obstacles. In various test terrains, it showed a markedly higher success rate compared to the single-network PPO method.

KEYWORDS

bipedal wheel-legged robot, curriculum learning, gradient conflict, mixture of experts, reinforcement learning

1 Introduction

The trade-off between wheeled efficiency and legged versatility is a significant issue in mobile robots. While wheeled robots excel on flat terrain and legged robots adapt to unstructured environments, bipedal wheel-legged robots have emerged as a promising hybrid to combine these advantages (Klemm et al., 2019; Wang et al., 2021; Guo et al., 2022; He et al., 2025; Wang et al., 2025). These systems can switch between continuous rolling for cruising and discrete stepping or jumping for obstacles (Klemm et al., 2020; Cui et al., 2021; Lee et al., 2024). However, unlocking this potential requires a control policy capable of mastering these distinct modes simultaneously. Traditional control of wheel-legged robots often relies on model-based optimization. For example, analytical second-order derivatives of rigid-body contact dynamics have been derived to enhance the multi-shooting Differential Dynamic Programming (DDP) algorithm (Singh et al., n.d.), thereby enabling a highly efficient approach to handling complex contact interactions in humanoid and hybrid locomotion robots. While such trajectory optimization methods deliver precise control performance based on analytical gradients, reinforcement learning (RL) provides an alternative by learning robust policies through trial and error, which can be more adaptable to unstructured environments.



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通过稀疏专家混合深度强化学习实现双足轮腿机器人的自适应多模式运动

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双足轮腿机器人结合了轮式运动的高能效与腿部运动的地形适应性。然而，在统一控制框架内实现这两种异质运动模式之间的平滑过渡仍然具有挑战性。本研究提出了一种集成专家混合（MoE）架构的强化学习控制框架。该方法采用“分而治之”策略，通过引入动态门控网络和Top-K稀疏激活机制，自动将不同运动模式分配给特定的专家子网络，有效解耦了梯度冲突。仿真结果表明，与单网络PPO方法相比，MoE增强算法在训练稳定性和奖励方面表现出显著提升。学习策略成功实现了平坦表面上的平滑滚动，并在遇到障碍物时过渡到动态抬腿步态。在各种测试地形中，其成功率明显高于单网络PPO方法。

关键词

双足轮腿机器人，课程学习，梯度冲突，专家混合，强化学习

1 引言

轮式效率与腿式多功能性之间的权衡是移动机器人领域的一个重要问题。轮式机器人在平坦地形上表现出色，而腿式机器人则能适应非结构化环境，双足轮腿机器人作为一种有前景的混合方案应运而生，旨在结合两者的优势（Klemm 等人，2019；Wang 等人，2021；Guo 等人，2022；He 等人，2025；Wang 等人，2025）。这些系统可以在连续滚动巡航与离散踏步或跳跃越障之间切换（Klemm 等人，2020；Cui 等人，2021；Lee 等人，2024）。然而，要释放这一潜力，需要一种能够同时掌握这些不同模式的控制策略。传统的轮腿机器人控制通常依赖于基于模型的优化。例如，研究人员推导了刚体接触动力学的解析二阶导数，以增强多重打靶微分动态规划（DDP）算法（Singh 等人，无日期），从而为处理人形和混合运动机器人中的复杂接触交互提供了一种高效方法。虽然这种轨迹优化方法基于解析梯度能够实现精确的控制性能，但强化学习（RL）提供了一种替代方案，通过试错学习鲁棒策略，能够更好地适应非结构化环境。

Developing a unified control policy is challenging due to the heterogeneous dynamics between high-precision rolling (Bouton et al., 2020; García and Duarte, 2024) and high-frequency maneuvers like jumping (Hwangbo et al., 2019; Lee et al., 2020; Zheng et al., 2025). Attempting to master these diverse skills within a monolithic reinforcement learning agent often triggers severe gradient conflicts (Yu et al., 2020), where updates for one mode interfere with the optimization of another. Consequently, the agent settles for a compromised policy, suffering from catastrophic forgetting or unstable convergence (Kirkpatrick et al., 2017; Schaul et al., 2019).

To address these limitations, this research proposes a Mixture of Experts (MoE) architecture integrated into the PPO framework, which effectively tackles the problem of gradient conflict (Shazeer et al., 2017; Celik et al., 2024; Li et al., 2024; Obando-Ceron et al., 2024; Akroure et al., 2022). The core concept of our approach is “divide-and-conquer.” Instead of forcing a single neural network to encode all locomotion primitives, we employ a dynamic gating mechanism that selects specific expert sub-networks based on the robot’s state. This architecture effectively disentangles the conflicting gradients. One expert can specialize in steady-state rolling, while another focuses on dynamic leg-lifting, with the gating network learning the optimal switching logic. Aside from innovations in algorithm architecture, we also propose a phased curriculum learning strategy (Kumar et al., 2021; Lee et al., 2024). The training process is divided into two phases. In the first phase, the agent is trained on specific terrains, focusing the acquisition of vertical mobility skills. In the second phase, the policy is generalized across randomized mixed terrains. Blind locomotion is realized by relying solely on proprioceptive history without privileged terrain information.

Based on the Diablo bipedal wheel-legged robot platform, we demonstrated the superiority of the proposed MoE-enhanced framework over the single-network PPO method. The results highlight three key contributions:

- **Training Stability:** A reinforcement learning framework integrated with the Mixture of Experts (MoE) architecture is introduced, which effectively resolves gradient conflicts in multi-modal locomotion tasks, significantly improving training stability and mitigating catastrophic forgetting compared to single-network PPO.
- **Higher Performance:** A dynamic gating network coupled with a Top-K sparse activation mechanism is introduced. This design automatically decouples heterogeneous motion modes, enabling expert subnet specialization and achieving higher reward peaks for better stability-agility balance.
- **Enhanced Traversability:** A two-phase curriculum learning strategy is designed to progress from deterministic terrains to noisy complex environments. This approach ensures robust generalization, granting the robot enhanced traversability over unstructured obstacles.

This work offers a novel solution for general motion learning in bipedal wheel-legged robots, providing a scalable pathway for deploying agile hybrid locomotion in complex environments.

2 Methods

2.1 Overall control framework

The robotic platform utilized in this study is the bipedal wheel-legged robot, Diablo (Direct Drive Tech, China), with simulations conducted in the Isaac Gym environment (Hwangbo et al., 2019; Lee et al., 2020; Kumar et al., 2021; Nahrendra et al., 2023). The overall control framework comprises a high-level decision controller and a low-level PD controller (Figure 1A). The high-level decision controller computes the target positions for each joint of the robot at a given time, which are then be scaled and transmitted to the low-level PD controller. The PD controller calculates the joint torques required to track these target positions, thereby driving the physics simulation.

2.2 Design of the controller based on the PPO algorithm

We initially trained the robot using a standard reinforcement learning approach. The objective was to achieve multi-gait capabilities within a single decision-making network. In this section, we employ the PPO algorithm (Schulman et al., 2017) to train the robot, and both the Actor and Critic consist of a neural network with dimension of [512, 256, 128].

2.2.1 Simulation environment setup

The training environment was constructed on the NVIDIA Isaac Gym simulation platform, utilizing an NVIDIA GeForce RTX 3090 Ti graphics card and configured with 4,096 parallel environments to efficiently train the Diablo robot, with a maximum episode duration of 20 s. The environment features specific terrains, which compel the robot to transition from rolling to stepping.

2.2.2 State space

In this study, the state space S is constructed as a high-dimensional composite vector of dimension R^{153} . The goal is to spatialize the temporal information, enabling a standard multi-layer perceptron (MLP) network to effectively perceive dynamic trends in the system, such as changes in acceleration and contact states. This vector is formed by concatenating real-time proprioceptive observations with historical joint data. Crucially, to ensure a truly blind locomotion policy, the input state S relies strictly on onboard proprioceptive sensors (IMU, joint encoders, and command input) and a history sequence of the most recent 10 frames. We explicitly exclude any exteroceptive data or privileged terrain heightmaps from the policy input. This configuration is maintained consistently across both the training and testing phases to validate the robot’s adaptability using only proprioception.

For the proprioceptive observations O_{base} , this component is represented as a vector in R^{33} , reflecting the robot’s current physical state and the task objectives. As shown in Equation 1, the specific elements include: the linear velocity of the robot’s base in the body coordinate system $v_t \in R^3$; the angular velocity $w_t \in R^3$; the projection of the gravity vector onto the body axes $g_t^{proj} \in R^3$; the commands $c_t \in R^6$, which encompass the target linear velocities v_x ,

由于高精度滚动 (Bouton 等人, 2020; García 和 Duarte, 2024) 与跳跃等高频机动 (Hwangbo 等人, 2019; Lee 等人, 2020; Zheng 等人, 2025) 之间存在异构动力学, 开发统一控制策略颇具挑战。试图在单体强化学习智能体内掌握这些多样化技能, 往往会引发严重的梯度冲突 (Yu 等人, 2020), 即一种模式的更新会干扰另一种模式的优化。因此, 智能体只能退而求其次, 采用折中策略, 并遭受灾难性遗忘或不稳定收敛的困扰 (Kirkpatrick 等人, 2017; Schaul 等人, 2019)。

为了解决这些局限性, 本研究提出了一种集成到PPO框架中的专家混合 (MoE) 架构, 有效解决了梯度冲突问题 (Shazeer 等人, 2017; Celik 等人, 2024; Li 等人, 2024; Obando-Ceron 等人, 2024; Akroure 等人, 2022)。我们方法的核心概念是“分而治之”。我们不是强迫单一神经网络编码所有运动基元, 而是采用动态门控机制, 根据机器人的状态选择特定的专家子网络。这种架构有效地解耦了冲突梯度。一个专家可以专门负责稳态滚动, 而另一个专家专注于动态抬腿, 门控网络学习最优的切换逻辑。除了算法架构的创新外, 我们还提出了一种分阶段课程学习策略 (Kumar 等人, 2021; Lee 等人, 2024)。训练过程分为两个阶段。在第一阶段, 智能体在特定地形上进行训练, 专注于垂直机动技能的获取。在第二阶段, 策略在随机混合地形上进行泛化。盲态运动通过仅仅依赖本体感觉历史来实现, 无需特权地形信息。

基于Diablo双足轮腿机器人平台, 我们验证了所提出的MoE增强框架相较于单网络PPO方法的优越性。实验结果凸显了三大关键贡献:

- **训练稳定性:** 引入了一种集成专家混合 (MoE) 架构的强化学习框架, 该框架有效解决了多模态运动任务中的梯度冲突问题, 与单网络PPO相比, 显著提升了训练稳定性并缓解了灾难性遗忘。

- **更高性能:** 引入了动态门控网络与Top-K稀疏激活机制相结合的设计。该设计能够自动解耦异构运动模式, 实现专家子网特化, 并获得更高的奖励峰值, 以实现更好的稳定性-敏捷性平衡。

- **增强越障能力:** 设计了一种两阶段课程学习策略, 从确定性地形逐步过渡到噪声复杂环境。该方法确保了鲁棒泛化能力, 使机器人在非结构化障碍物上具备更强的越障能力。

这项作为双足轮腿机器人的通用运动学习提供了一种新颖的解决方案, 为在复杂环境中部署敏捷混合运动提供了一条可扩展的路径。

2 方法

2.1 总体控制框架

本研究所使用的机器人平台是双足轮腿机器人Diablo (中国Direct Drive Tech公司), 仿真在Isaac Gym环境中进行 (Hwangbo 等人, 2019; Lee 等人, 2020; Kumar 等人, 2021; Nahrendra 等人, 2023)。整体控制框架包括一个高层决策控制器和一个底层PD控制器 (图1A)。高层决策控制器计算机器人各关节在给定时间的目标位置, 这些目标位置随后被缩放并传输至底层PD控制器。PD控制器计算跟踪这些目标位置所需的关节力矩, 从而驱动物理仿真。

2.2 基于PPO算法的控制器设计

我们最初使用标准强化学习方法训练机器人, 目标是在单一决策网络中实现多步态能力。在本节中, 我们采用PPO算法 (Schulman 等人, 2017) 来训练机器人, Actor和Critic均由维度为[512, 256, 128]的神经网络组成。

2.2.1 仿真环境设置

训练环境构建于NVIDIA Isaac Gym仿真平台上, 使用NVIDIA GeForce RTX 3090 Ti显卡, 并配置了4,096个并行环境以高效训练Diablo机器人, 最大回合时长为20秒。该环境包含特定的地形, 迫使机器人从滚动状态过渡到行走状态。

2.2.2 状态空间

在本研究中, 状态空间 S 被构建为维度为 R^{153} 的高维复合向量。其目标是将时间信息空间化, 使标准多层感知机 (MLP) 网络能够有效感知系统的动态趋势, 如加速度变化和接触状态变化。该向量由实时本体感觉观测与历史关节数据拼接而成。至关重要, 为确保真正的盲行走策略, 输入状态 S 严格依赖于机载本体感觉传感器 (IMU、关节编码器和指令输入) 以及最近10帧的历史序列。我们明确将任何外部感觉数据或特权地形高度图排除在策略输入之外。该配置在训练和测试阶段保持一致, 以验证机器人仅利用本体感觉的适应性。

对于本体感觉观测 O_{base} , 该组件表示为 R^{33} 中的向量, 反映机器人当前的物理状态和任务目标。如公式1所示, 具体元素包括: 机体坐标系中机器人基座的线速度 $v_t \in R^3$; 角速度 $w_t \in R^3$; 重力向量在机体轴上的投影 $g_t^{proj} \in R^3$; 指令 $c_t \in R^6$, 其中包括目标线速度 v_x ,

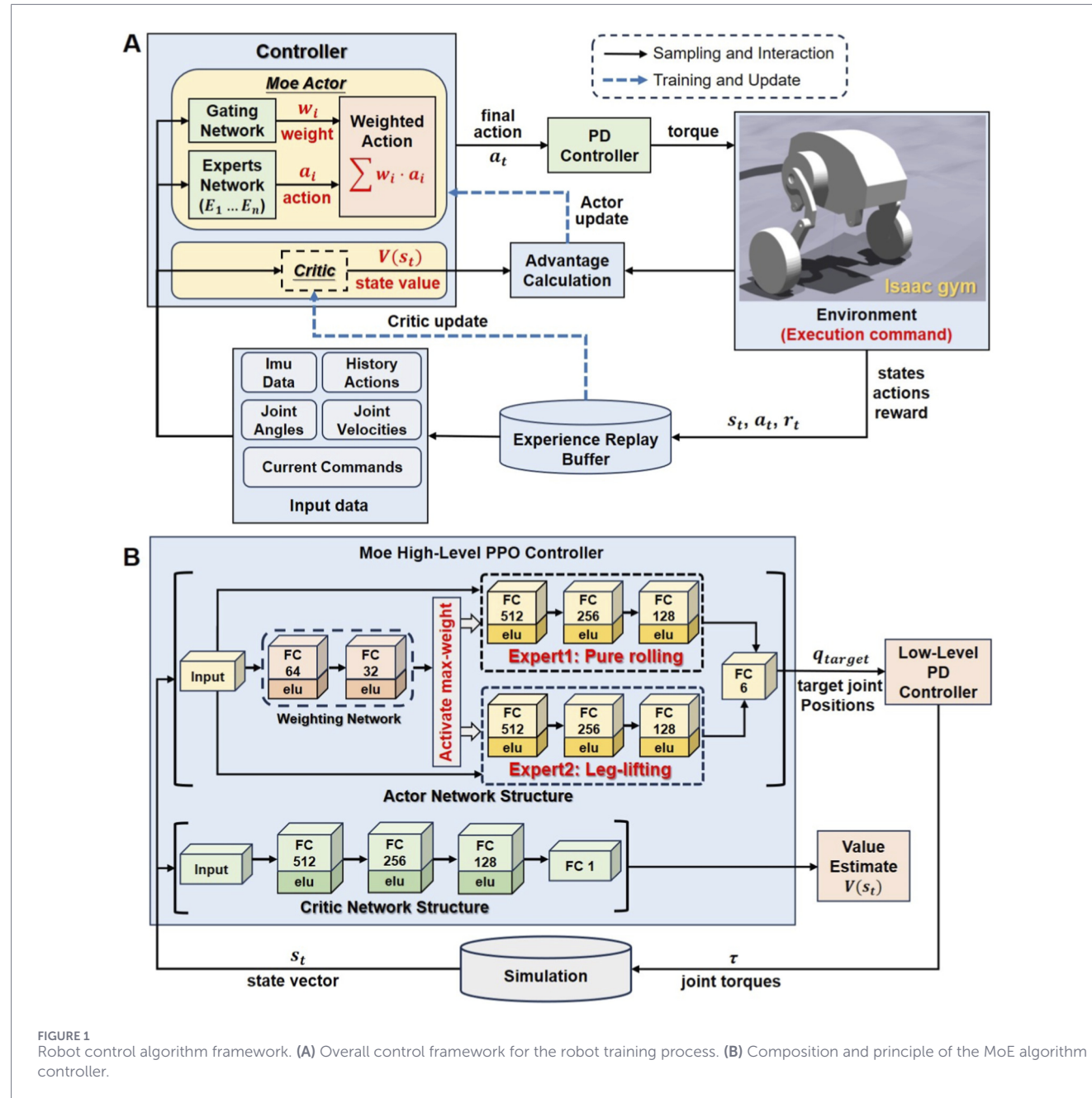


FIGURE 1 Robot control algorithm framework. (A) Overall control framework for the robot training process. (B) Composition and principle of the MoE algorithm controller.

v_y and yaw rate w_{yaw} , as well as the jump height target h_{jump} and the adjustment angles for the left and right legs θ_{left} , θ_{right} ; the current position error for the joints $e_t = (q_t - q_{default}) \in R^6$; the joint velocities $\dot{q}_t \in R^6$; and the action vector from the previous time step $a_{t-1} \in R^6$.

$$O_{base} = \langle v_t, w_t, g_j^{proj}, c_t, e_t, \dot{q}_t, a_t \rangle \quad (1)$$

The historical observations O_{hist} is represented as a vector in R^{120} , which stores the joint data from the most recent 10 frames, corresponding to a continuous segment of time in the physical simulation. This includes the historical position errors of the robot's six joints $h_{pos} = [e_{t-10}, \dots, e_{t-1}] \in R^{60}$ and the historical velocities $h_{vel} = [\dot{q}_{t-10}, \dots, \dot{q}_{t-1}] \in R^{60}$. At any given time t , the input state vector

is formulated as follows:

$$S_t = \langle O_{base}, h_{pos}, h_{vel} \rangle \quad (2)$$

2.2.3 Action space

In this study, we employ a continuous action space $A \in R^6$ for the precise control of the six degrees of freedom on both sides of the bipedal wheel-legged robot. The policy network does not directly output torques. Instead, it generates a normalized position deviation vector $a_t \in R^6$, which is transformed into target joint positions q_t^* through linear mapping, as shown in Equation 3. Here, q_{nom} represents the robot's initial posture, and σ is the action scaling

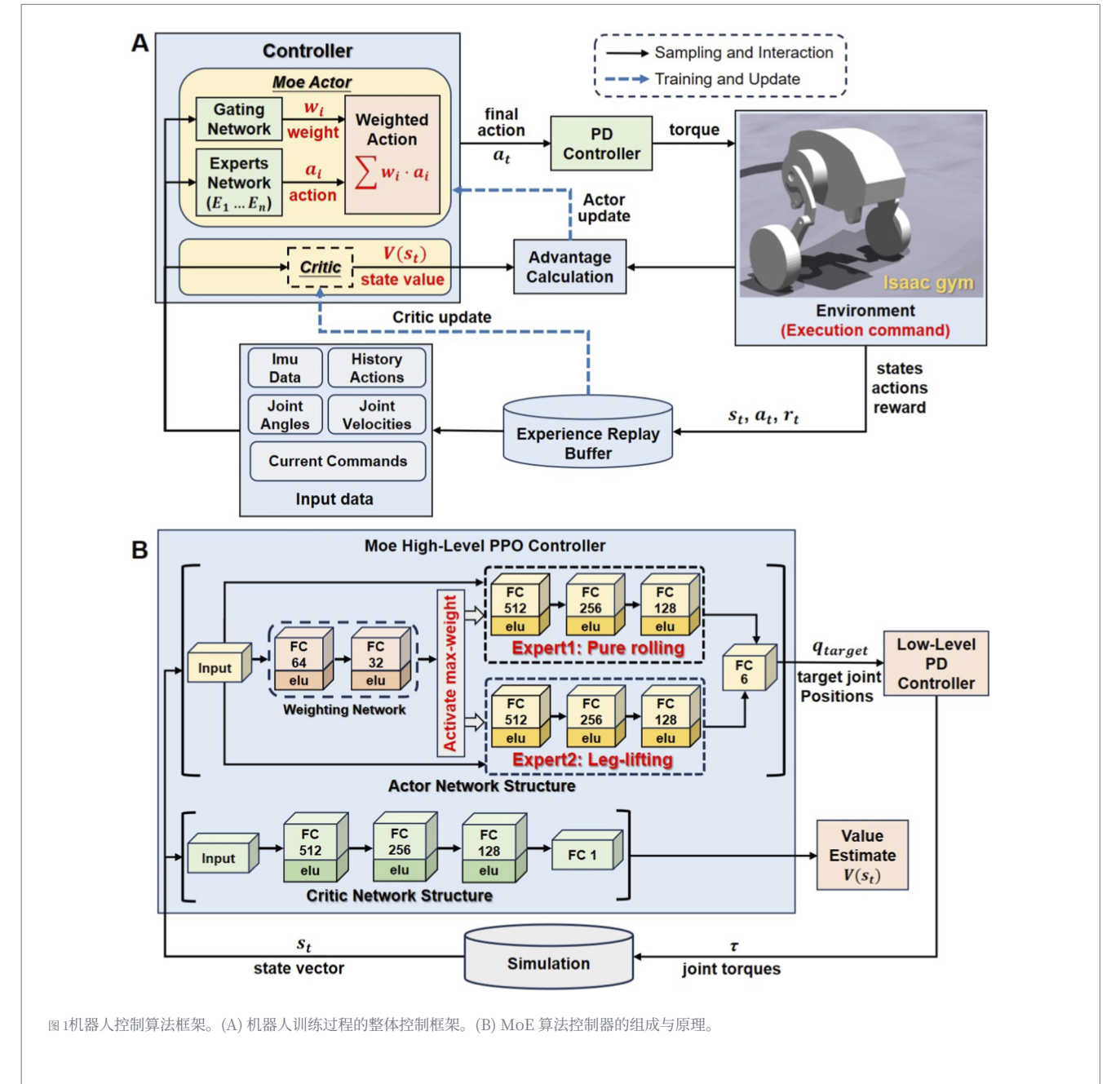


图1机器人控制算法框架。(A) 机器人训练过程的整体控制框架。(B) MoE 算法控制器的组成与原理。

v_y 和偏航角速度 w_{yaw} , 以及跳跃高度目标 h_{jump} 和左右腿的调整角度 θ_{left} , θ_{right} ; 关节的当前位置误差 $e_t = (q_t - q_{default}) \in R^6$; 关节速度 $\dot{q}_t \in R^6$; 以及来自上一个时间步的动作向量 $a_{t-1} \in R^6$ 。

的公式如下:

$$S_t = \langle O_{base}, h_{pos}, h_{vel} \rangle \quad (2)$$

2.2.3 动作空间

在本研究中, 我们采用连续动作空间 $A \in R^6$ 来精确控制双足轮腿机器人两侧的六自由度。策略网络不直接输出力矩, 而是生成一个归一化位置偏差向量 $a_t \in R^6$, 该向量通过线性映射转换为目标关节位置 q_t^* , 如公式3所示。此处, q_{nom} 表示机器人的初始姿态, 而 σ 是动作缩放

$$O_{base} = \langle v_t, w_t, g_j^{proj}, c_t, e_t, \dot{q}_t, a_t \rangle \quad (1)$$

历史观测 O_{hist} 表示为向量 R^{120} , 其中存储了最近10帧的关节数据, 对应物理仿真中的连续时间段。这包括机器人六个关节的历史位置误差 $h_{pos} = [e_{t-10}, \dots, e_{t-1}] \in R^{60}$ 以及历史速度 $h_{vel} = [\dot{q}_{t-10}, \dots, \dot{q}_{t-1}] \in R^{60}$ 。在任意给定时间 t , 输入状态向量

TABLE 1 Specific reward parameters and calculation methods.

Category	Reward item name	Scale	Calculation method
Task objective rewards	Tracking lin vel	5.5	$r_{lin_vel} = \exp\left(-\left\ v_{xy} - v_{xy}^{com}\right\ ^2 / \sigma_{lin}\right)$
	Tracking ang vel	5.5	$r_{ang_vel} = \exp\left(-\left(w_{yaw} - w_{yaw}^{com}\right)^2 / \sigma_{ang}\right)$
	Base height	−0.25	$r_{height} = -\left(h_{base} - h_{target}^{com}\right)^2$
Gait induction and constraint reward	Leg lifting	20	$r_{leg_lift} = mean(z_{wheel-L} - z_{wheel-R})$
	No fly	0.25	$r_{no_fly} = 1(if \sum contact = = 2)$
	No moonwalk	−2.5	$r_{no_moonwalk} = -\left(P_{proj_L} + P_{proj_R}\right)^2$
Stability and physical constraint reward	Orientation	−5.5	$r_{orientation} = \left\ \xi_{xy}^{proj}\right\ $
	Lin vel z	−0.1	$r_{lin_vel_z} = v_z^2$
	Ang vel xy	−0.25	$r_{ang_vel_xy} = w_x^2 + w_y^2$
Motion smoothness and safety reward	Collision	−180	$r_{collision} = -1(if bady_contact > 0.1)$
	Dof pos limits	−0.5	$r_{dof_limit} = -\sum clip(q - q_{limit})$
	Action rate	−0.008	$r_{action_rate} = -\left\ a_t - a_{t-1}\right\ ^2$
	Torques	-1e-5	$r_{torque} = -\left\ \tau\right\ ^2$

factor. These high-level position commands are then passed to a low-level PD controller to compute the final execution torques τ_t , with the calculation method outlined in Equation 4. In this context, q_t and $\dot{q}_t$ denote the current joint positions and velocities, while K_p and K_d represent the specified position stiffness and velocity damping gains.

$$q_t^* = q_{nom} + \sigma \cdot a_t$$

$$\tau_t = clip\left(K_p(q_t^* - q_t) - K_d\dot{q}_t, -\tau_{max}, \tau_{max}\right)$$

2.2.4 Reward function

The reward function R consists of four components: task objective rewards, gait induction and constraint rewards, stability and physical constraint rewards, and motion smoothness and safety rewards. The specific parameters and calculation methods for these reward components are detailed in Table 1. In the gait induction and constraint rewards, it is worth noting that the “leg lifting reward” and “no moonwalk reward” are complementary in their functionality. Specifically, the positive “leg lifting reward” is designed to induce a vertical height difference between the hub centers of the robot’s left and right legs, thereby prompting a leg-lifting and stepping gait while preventing the training process from converging to the local optimum of pure rolling locomotion. In contrast, the negative “no moonwalk reward” is introduced to eliminate horizontal offset between the left and right leg hub centers, effectively suppressing the undesired gait of staggered wheel positions along the forward direction.

2.2.5 Domain randomization and termination conditions

To enhance the robustness of the policy, this study implements a comprehensive domain randomization framework that includes

environmental dynamics randomization, initial state perturbations, sensor noise injection, and external mechanical impacts. The ground friction coefficient μ of the physics engine is independently sampled from a uniform distribution $U(0.5, 1.25)$ during each reset, and the base’s horizontal position is randomly offset within a range of ± 1 m. The observation space is augmented with parameterized additive noise, as shown in Equation 5, where $n \sim U(-s, s)$, and the noise amplitude s is set with differentiated intensities, the specific values are given in Supplementary Table S4 of the Supplementary Material. Furthermore, a random horizontal velocity impact of 1.0 m/s is applied every 15 s to enforce balance recovery. The termination condition d_t is triggered by both time truncation and physical contact.

$$O_{noisy} = O_{origin} + n$$

2.3 Design of the MoE controller

Based on the original PPO framework, we introduced a MoE mechanism in the Actor. The algorithm is illustrated in Figure 1B. The new algorithm maintains consistency with the original PPO algorithm in terms of state space, action space, and reward function. For the new algorithm that incorporates the MoE, the Actor is composed of a gating network with dimensions of [128, 64] and two expert networks sized [512, 256, 128]. At each time step, the final action output of the Actor corresponds to the expert network with the highest weight.

2.3.1 Composition of the loss function

The loss function L_{base} of the basic PPO algorithm is shown in Equation 6. Here, $L^{clip}(\theta)$ denotes the clipped surrogate loss that restricts the magnitude of policy updates, $L^{value}(\theta)$ represents the squared error loss for evaluating state values, and $H[\pi_\theta]$ is a linear combination of entropy to encourage policy

表 1 具体奖励参数及计算方法。

类别	奖励项 name	尺度	计算方法
任务目标奖励	跟踪线速度	5.5	$r_{lin_vel} = \exp(-\ v_{xy} - v_{xy}^{com}\ ^2 / \sigma_{lin})$
	跟踪角速度	5.5	$r_{ang_vel} = \exp(-(w_{yaw} - w_{yaw}^{com})^2 / \sigma_{ang})$
	基座高度	−0.25	$r^{height} = -\left(h_{base} - h_{参数目标}^{com}\right)^2$
步态诱导与约束奖励	抬腿	20	$r_{leg_lift} = mean(z_{wheel-L} - z_{wheel-R})$
	禁止飞行	0.25	$r_{no_fly} = 1(if \sum 接触 = = 2)$
	禁止月球漫步	−2.5	$r_{no_moonwalk} = -\left(P_{proj_L} + P_{proj_R}\right)^2$
稳定性与物理约束奖励	方向	−5.5	$r_{orientation} = \left\ g_{xy}^{proj}\right\ $
	线速度z	−0.1	$r_{lin_vel_z} = v_z^2$
	角速度xy	−0.25	$r_{ang_vel_xy} = w_x^2 + w_y^2$
运动平滑性与安全性奖励	碰撞	−180	$r_{collision} = -1(if bady_接触 > 0.1)$
	自由度位置限制	−0.5	$r_{dof_limit} = -\sum clip(q - q_{limit})$
	动作速率	−0.008	$r_{action_rate} = -\left\ a_t - a^{t-1}\right\ ^2$
	力矩	-1e-5	$r_{torque} = -\left\ \tau\right\ ^2$

因子。这些高层位置指令随后传递给低层PD控制器，以计算最终的执行力矩 τ_t ，计算方法详见公式4。在此上下文中， q_t 和 q_t 分别表示当前关节位置和速度，而 K_p 和 K_d 代表指定的位置刚度和速度阻尼增益。

$$q_t^* = q_{nom} + \sigma \cdot a_t$$

$$\tau_t = clip\left(K_p(q_t^* - q_t) - K_d\dot{q}_t, -\tau_{max}, \tau_{max}\right)$$

2.2.4 奖励函数

奖励函数 R 由四个部分组成：任务目标奖励、步态诱导与约束奖励、稳定性与物理约束奖励，以及运动平滑性与安全性奖励。这些奖励组件的具体参数和计算方法详见表 1。在步态诱导与约束奖励中，值得注意的是，“抬腿奖励”和“无月球漫步奖励”在功能上是互补的。具体而言，正向的“抬腿奖励”旨在诱导机器人左右腿轮毂中心之间产生垂直高度差，从而促使形成抬腿迈步步态，同时防止训练过程收敛到纯滚动运动的局部最优解。相反，负向的“无月球漫步奖励”则用于消除左右腿轮毂中心之间的水平偏移，有效抑制沿前进方向出现交错轮位的不良步态。

2.2.5 域随机化与终止条件

为增强策略的鲁棒性，本研究实施了一个全面的域随机化框架，包括

环境动力学随机化、初始状态扰动、传感器噪声注入和外部机械冲击。物理引擎的地面摩擦系数 μ 在每次重置时独立地从均匀分布 $U(0.5, 1.25)$ 中采样，且基座的水平位置在 ± 1 m 范围内随机偏移。观测空间增加了参数化的加性噪声，如 公式 5 所示，其中 $n \sim U(-s, s)$ ，噪声幅度 s 设置了差异化强度，具体数值见 补充表S4 中的 补充材料。此外，每15秒施加一次 1.0 m/s 的随机水平速度冲击，以强制实现平衡恢复。终止条件 d_t 由时间截断和物理接触共同触发。

$$O_{noisy} = O_{origin} + n$$

2.3 MoE控制器的设计

基于原始PPO框架，我们在Actor中引入了MoE机制。该算法如图1B所示。新算法在状态空间、动作空间和奖励函数方面与原始PPO算法保持一致。对于融合了MoE的新算法，Actor由维度为[128, 64] 的门控网络和两个大小为 [512, 256, 128]的专家网络组成。在每个时间步，Actor的最终动作输出对应于权重最高的专家网络。

2.3.1 损失函数的组成

损失函数 L_{base} 基础PPO算法的损失函数如公式6所示。其中， $L^{clip}(\theta)$ 表示裁剪替代损失，用于限制策略更新的幅度， $L^{value}(\theta)$ 表示用于评估状态值的平方误差损失，而 $H[\pi_\theta]$ 是熵的线性组合，用于鼓励策略

exploration. The coefficients c_1 and c_2 are used to adjust the contributions of the latter two losses.

$$L_{base} = L^{clip}(\theta) + c_1 \cdot L^{value}(\theta) - c_2 \cdot H[\pi_\theta] \quad (6)$$

After introducing the MoE mechanism, we augmented the total loss function to L_{total} as shown in Equation 7 to prevent mode collapse within the expert networks. The newly added term is the load balancing auxiliary loss L_{aux} , which is regulated by the coefficient c_{moe} . The calculation method for this loss is detailed in Equation 8, where w_i denotes the average gating weight of the i -th expert in the current training batch, and N represents the total number of experts. This term enforces uniform activation and utilization of all experts by minimizing both the negative entropy of the average weights and the mean absolute deviation.

$$L_{base} = L_{base} + c_{moe} \cdot L_{aux} \quad (7)$$

$$L_{aux} = \sum_{i=1}^N w_i \ln w_i + 0.5 \cdot \frac{1}{N} \sum_{i=1}^N \left| w_i - \frac{1}{N} \right| \quad (8)$$

2.3.2 Top-K mechanism in the MoE framework

To achieve computational sparsity and encourage the policy network to transition clearly between different motion modes, we incorporate a Top-K gating mechanism in the policy framework (Chen et al., 2025; D'Souza et al., 2025; Huang et al., 2025; Xu et al., 2025; Vincze et al., 2025). For state input S_t , the gating network G_ϕ generates a probability distribution $g \in R^N$ for N experts via Softmax (Equation 9). A truncation operation then retains only the top k experts to form the active set K (Equation 10). Selected weights are re-normalized to ensure effective gradient propagation. The final output μ_t is a linear combination of the outputs $E_{\theta-i}(S_t)$ from these k active experts (Equation 11), enforcing sparse activation and allowing each expert to focus on specific sub-tasks. In this study, the gating network and expert networks take the same input S_t (Equation 2). Meanwhile, we fix the number of activated experts K to 1 at each time step, thereby ensuring that only the most relevant expert is active at any given time to maximize the decoupling of motion primitives.

$$g = \text{Softmax}(W_g \cdot s_t + b_g) \quad (9)$$

$$K = \{i | g_i \in \text{top-}k(g)\} \quad (10)$$

$$\mu_t = \sum_{i \in K} w_i \cdot E_{\theta_i}(S_t) \quad (11)$$

2.4 Phased training design

For a bipedal wheel-legged robot, the learning process begins with mastering fundamental equilibrium on planar surfaces, followed by tracking velocity commands via rolling locomotion. Subsequently, the robot negotiates slopes and overcomes vertical obstacles on complex terrains using leg-lifting gaits. Ultimately, it masters dynamically switching between rolling and leg-lifting modes. To this end, we

designed a progressive training framework to induce these skills sequentially. Central to this approach is the two-phase training pipeline illustrated in Figure 2A, where each phase features distinct terrain compositions and complexity levels tailored to specific skills. The first phase consists exclusively of flat ground, slopes, and vertical stairs. An inclination threshold is defined such that slopes exceeding this value are rendered as vertical stairs in the simulation. This process evolves the initial policy π_0 into an obstacle-capable policy π_1 . The second phase involves transferring the policy π_1 to a comprehensive terrain set for re-training. To increase terrain complexity, Phase 2 incorporates wave terrain and discrete terrain into the training set, with Perlin noise added to introduce irregular surface perturbations. This phase serves as a comprehensive training ground for hybrid locomotion switching and dynamic balancing. Building upon the results of Phase 1, this process culminates in the final generalized policy π_2 .

2.5 Curriculum learning design

Prior research demonstrates that curriculum learning—where task difficulty is incrementally increased—is a highly effective strategy for training complex locomotion policies. Accordingly, we propose an adaptive terrain curriculum designed to facilitate effective learning for bipedal wheel-legged robots in unstructured environments. At the onset of training, robots are uniformly distributed across the lowest difficulty tier of each terrain type, and the terrain difficulty level is dynamically updated based on the robot's real-time performance. If the robot successfully traverses more than half the terrain length within a fixed time, it is deemed to have mastered the current difficulty. Consequently, the terrain level is incremented upon the next reset. Conversely, the difficulty level is decremented, allowing the robot to consolidate basic skills on simpler terrain. To mitigate catastrophic forgetting, robots that reach the maximum difficulty level are randomly reassigned to intermediate terrains.

3 Result

3.1 Training performance

Since the Actor in MoE-enhanced algorithm contains two networks of size [512, 256, 128], its total parameter count is approximately twice that of the standard PPO baseline. To exclude the impact of increased network capacity, this study introduces an additional large PPO baseline with a similar parameter count to MoE-enhanced algorithm. The large PPO baseline has a network size of [768, 384, 192], corresponding to scaling each layer of the Actor in the standard PPO baseline by a factor of 1.5. In both training phases, all three algorithms are trained with identical hyperparameters, including reward term weights, terrain settings, randomization parameters, and total training iterations. In the first phase, no Perlin noise is applied to the terrain, while in the second phase, Perlin noise is added to introduce irregular surface disturbances, as shown in Figure 3A. Both phases contain 6,000 training iterations, and the average rewards are compared, with results presented in Figures 3B,C, respectively. In the first phase, no obvious training collapse is observed for the three algorithms. Before around 1,600 iterations, their reward values and convergence speeds are nearly identical. After 1,600 iterations, the reward curves

探索。系数 c_1 和 c_2 用于调整后两项损失的贡献。

$$L_{base} = L^{clip}(\theta) + c_1 \cdot L^{value}(\theta) - c_2 \cdot H[\pi_\theta] \quad (6)$$

在引入 MoE机制后，我们对总损失函数进行了扩展，得到 L_{total} ，如公式 7 所示，以防止专家网络内部出现模式坍塌。新增项为负载均衡辅助损失 L_{aux} ，由系数 c_{moe} 进行调节。该损失的计算方法详见公式 8，其中 w_i 表示当前训练批次中第 i 个专家的平均门控权重， N 表示专家总数。该项通过最小化平均权重的负熵和平均绝对偏差，强制所有专家被均匀激活和利用。

$$L_{base} = L_{base} + c_{moe} \cdot L_{aux} \quad (7)$$

$$L_{aux} = \sum_{i=1}^N w_i \ln w_i + 0.5 \cdot \frac{1}{N} \sum_{i=1}^N \left| w_i - \frac{1}{N} \right| \quad (8)$$

2.3.2 MoE 框架中的 Top-K 机制

为了实现计算稀疏性并促使策略网络在不同运动模式之间清晰切换，我们在策略框架中引入了 Top-K 门控机制（Chen 等人，2025；D' Souza 等人，2025；Huang 等人，2025；Xu 等人，2025；Vincze 等人，2025）。对于状态输入 S_t ，门控网络 G_ϕ 通过 Softmax 为 N 个专家生成概率分布 $g \in R^N$ （公式 9）。随后，截断操作仅保留前 k 个专家以构成激活集 K （公式 10）。选中的权重会被重新归一化，以确保有效的梯度传播。最终输出 μ_t 是这 k 个激活专家的输出 $E_{\theta-i}(S_t)$ 的线性组合（公式 11），从而实现稀疏激活，并允许每个专家专注于特定的子任务。在本研究中，门控网络和专家网络接收相同的输入 S_t （公式 2）。同时，我们将每个时间步激活的专家数量 K 固定为 1，从而确保在任何给定时刻只有最相关的专家处于激活状态，以最大化运动基元的解耦。

$$g = \text{Softmax}(W_g \cdot s_t + b_g) \quad (9)$$

$$K = \{i | g_i \in \text{top-}k(g)\} \quad (10)$$

$$\mu_t = \sum_{i \in K} w_i \cdot E_{\theta_i}(S_t) \quad (11)$$

2.4 分阶段训练设计

对于双足轮腿机器人，学习过程始于掌握平面上的基本平衡，随后通过滚动运动跟踪速度指令。接着，机器人利用抬腿步态在复杂地形上应对斜坡并克服垂直障碍物。最终，它掌握了在滚动和抬腿模式之间动态切换的能力。为此，我们

设计了一个渐进式训练框架，以按顺序引导这些技能。该方法核心理念是如图2A所示的两阶段训练流程图2A，其中每个阶段具有针对特定技能量身定制的地形组成和复杂度水平。第一阶段仅包含平地、斜坡和垂直楼梯。定义了一个倾斜阈值，使得超过该值的斜坡在仿真中被渲染为垂直楼梯。此过程将初始策略 π_0 演化为具备障碍物应对能力的策略 π_1 。第二阶段涉及将策略 π_1 迁移至综合地形集进行再训练。为增加地形复杂度，第二阶段将波浪地形和离散地形纳入训练集，并添加Perlin噪声以引入不规则表面扰动。该阶段作为混合运动切换和动态平衡的综合训练场。基于第一阶段的成果，此过程最终形成通用的最终策略 π_2 。

2.5 课程学习设计

先前研究表明，课程学习——即任务难度逐步递增——是训练复杂运动策略的一种高效策略。因此，我们提出了一种自适应地形课程，旨在促进双足轮腿机器人在非结构化环境中的有效学习。训练开始时，机器人均匀分布在每种地形类型的最低难度等级中，地形难度等级根据机器人的实时表现动态更新。如果机器人在固定时间内成功穿越超过一半的地形长度，则视为已掌握当前难度。因此，在下次重置时，地形等级会递增。相反，如果难度等级降低，机器人可以在较简单的地形上巩固基本技能。为了缓解灾难性遗忘，达到最大难度等级的机器人会被随机重新分配到中等难度的地形中。

3 结果

3.1 训练表现

由于MoE增强算法中的Actor包含两个大小为 [512, 256, 128]，的网络，其总参数量约为标准PPO基线的两倍。为排除网络容量增加的影响，本研究引入了一个与MoE增强算法参数量相近的大型PPO基线。大型PPO基线的网络大小为 [768, 384, 192]，，相当于将标准PPO基线中Actor的每一层按1.5倍比例缩放。在两个训练阶段中，三种算法均使用相同的超参数进行训练，包括奖励项权重、地形设置、随机化参数和总训练迭代次数。第一阶段中，地形不施加Perlin噪声；第二阶段中，则添加Perlin噪声以引入不规则表面扰动，如图3A所示。两个阶段均包含6,000次训练迭代，并比较平均奖励，结果分别如图3B、C所示。在第一阶段，三种算法均未出现明显的训练崩溃。在约1,600次迭代之前，它们的奖励值和收敛速度几乎相同。1,600次迭代之后，奖励曲线

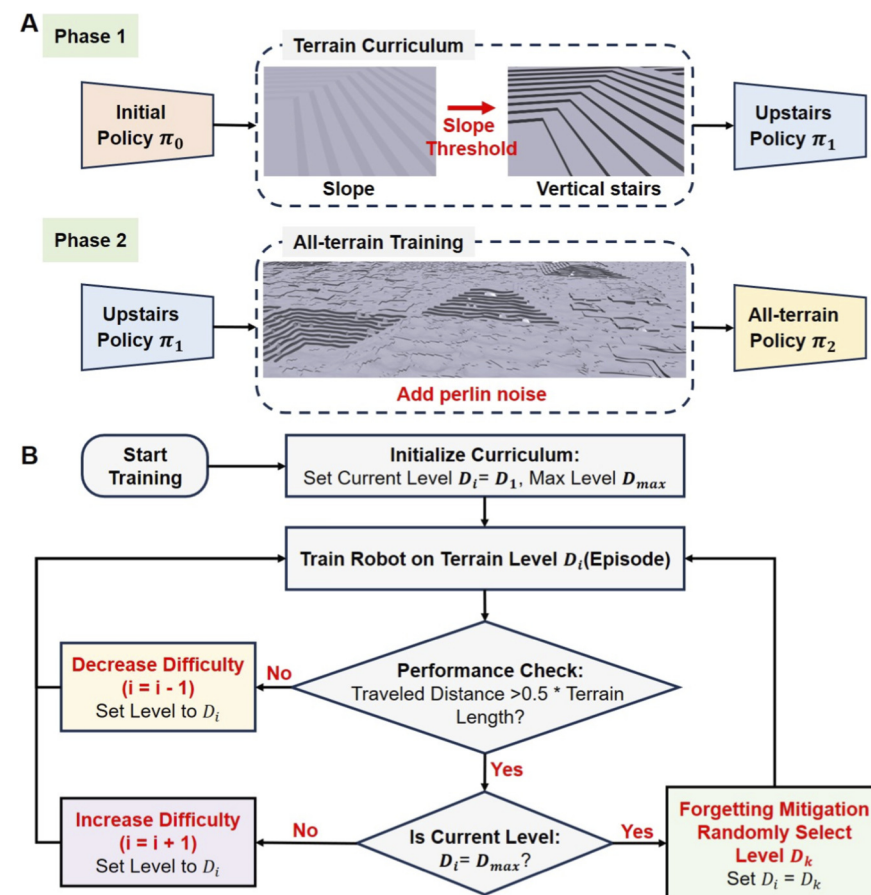


FIGURE 2
Schematics of the training process. (A) Schematic of the phased training design. (B) Schematic of the curriculum learning design.

diverge significantly: the reward of the MoE-enhanced algorithm rises steadily, while that of the standard PPO baseline declines continuously. Although the reward of the large PPO baseline does not drop obviously, it shows no upward trend. By the end of the first phase, the reward of the MoE-enhanced algorithm is significantly higher than the other two baselines. Initially, the terrain mainly includes flat ground, slopes, and low stairs with small differences in dynamic properties. For low stairs, the robot can even traverse them via pure rolling, with dynamics similar to normal flat-ground locomotion and slope traversal. As the training progressed and stair heights increased, reaching a maximum height of 14 cm, which is much larger than the wheel radius, the robot can only overcome such obstacles through jumping or leg-lifting stepping, whose dynamics differ greatly from pure rolling. Training with a single network easily causes gradient update interference: after learning leg-lifting or jumping, the robot partially forgets pure rolling; similarly, improving pure rolling leads to partial forgetting of obstacle-crossing skills. Thus, the policy learned by a single network is a compromised solution. When facing higher stairs, its performance is insufficient, leading to stagnant or even severely degraded rewards.

In the second phase, Perlin noise with an amplitude of 6 cm is introduced into the terrain, and terrain types are extended to full terrain. Under these conditions, both the standard PPO baseline and large PPO baseline exhibit severe training instability:

reward curves fail to converge, and training collapse frequently occurs. The primary cause is that Perlin noise increases terrain surface unevenness, leading to large fluctuations in the robot's state observations at each time step. The policy continuously chases changing optimal behavior, resulting in reward volatility. If the frequency of terrain variations exceeds the adaptation speed of the policy, training collapse occurs. For single-network algorithms, limited learning capacity causes a trade-off dilemma, making stable training infeasible on rough terrain and preventing a robust final policy. In contrast, the MoE-enhanced algorithm maintains favorable reward curves throughout entire training process. In the first phase, its policy gradually adapts to increasing stair heights, with consistent reward rises. In the second phase, since the two expert networks' skills were specialized in the first phase, with each expert responsible for distinct gait tasks, stable locomotion can be achieved by alternating between different experts even on rough terrain surfaces, thereby ensuring training stability. To further verify MoE-enhanced algorithm stability, we trained all three algorithms with different random seeds. The results are shown in [Supplementary Figure S1](#) of the [Supplementary Material](#). The results demonstrate that the MoE-enhanced algorithm performs stably across different seeds, whereas the single-network standard PPO and large PPO baselines exhibit significant variations across seeds and struggle to learn a stable policy. Overall, these results preliminarily

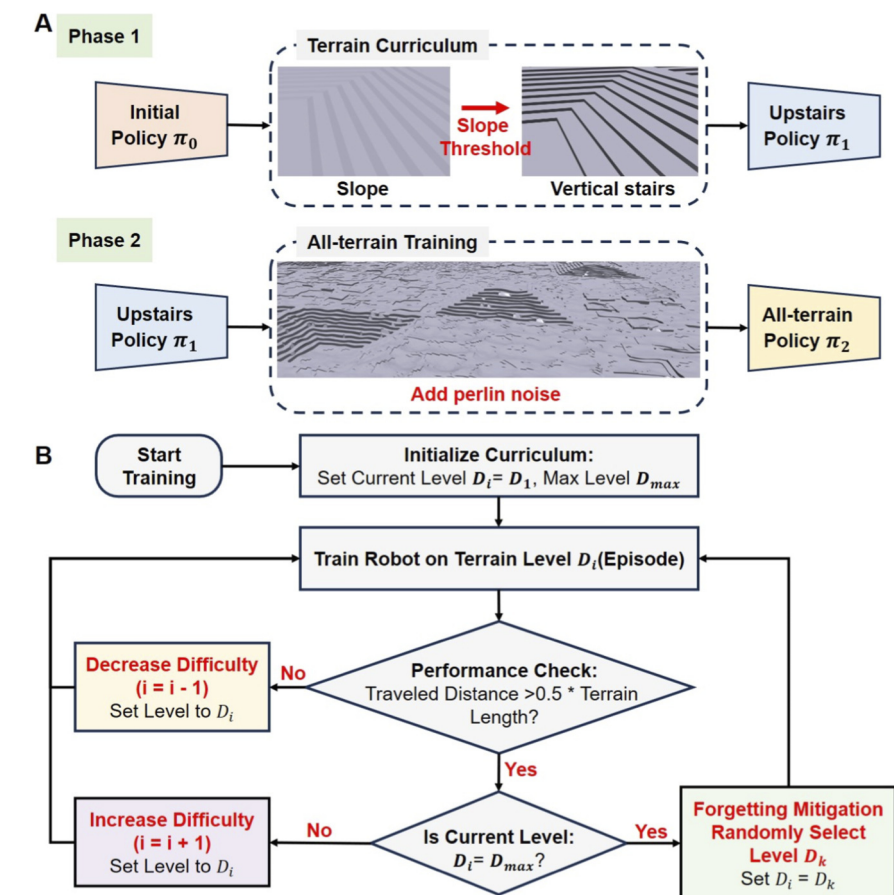


FIGURE 2
S训练过程示意图。(A) 分阶段训练设计示意图。(B) 课程学习设计示意图

n.

出现显著分化: MoE增强算法的奖励稳步上升, 而标准PPO基线的奖励持续下降。尽管大型PPO基线的奖励没有明显下降, 但也未呈现上升趋势。第一阶段结束时, MoE增强算法的奖励显著高于其他两个基线。初始阶段, 地形主要包括平地、斜坡和低台阶, 其动力学特性差异较小。对于低台阶, 机器人甚至可以通过纯滚动方式通过, 其动力学与正常的平地运动和斜坡穿越相似。随着训练推进, 台阶高度逐渐增加, 最高达到14厘米, 远大于车轮半径, 此时机器人只能通过跳跃或抬腿踏步来克服此类障碍, 其动力学特性与纯滚动差异巨大。使用单一网络进行训练容易导致梯度更新干扰: 在学习抬腿踏步或跳跃后, 机器人会部分遗忘纯滚动技能; 同样, 改进纯滚动也会导致越障技能的部分遗忘。因此, 单一网络学习到的策略是一种折中方案。面对更高台阶时, 其表现不足, 导致奖励停滞甚至严重退化。

在第二阶段, 将振幅为 6 cm 的 Perlin 噪声引入地形中, 并将地形类型扩展至全地形。在这些条件下, 标准PPO基线和大型PPO基线均表现出严重的训练不稳定性:

奖励曲线无法收敛, 训练崩溃频繁发生。主要原因是Perlin噪声增加了地形表面不平整度, 导致机器人在每个时间步的状态观测出现较大波动。策略不断追逐变化的最优行为, 导致奖励波动。如果地形变化的频率超过策略的适应速度, 就会发生训练崩溃。对于单网络算法, 有限的学习能力导致权衡困境, 使其在崎岖地形上无法进行稳定训练, 也无法获得稳健的最终策略。相比之下, MoE增强算法在整个训练过程中保持了良好的奖励曲线。在第一阶段, 其策略逐渐适应不断增加的台阶高度, 奖励持续上升。在第二阶段, 由于两个专家网络的技能在第一阶段已专门化, 每个专家负责不同的步态任务, 即使在崎岖地形表面上, 也可以通过在不同专家之间交替来实现稳定运动, 从而确保训练稳定性。为进一步验证MoE增强算法的稳定性, 我们使用不同的随机种子训练了所有三种算法。结果见[补充图S1](#)中的[补充材料](#)。结果表明, MoE增强算法在不同种子下表现稳定, 而单网络标准PPO和大型PPO基线在不同种子间表现出显著差异, 且难以学习到稳定的策略。总体而言, 这些结果初步

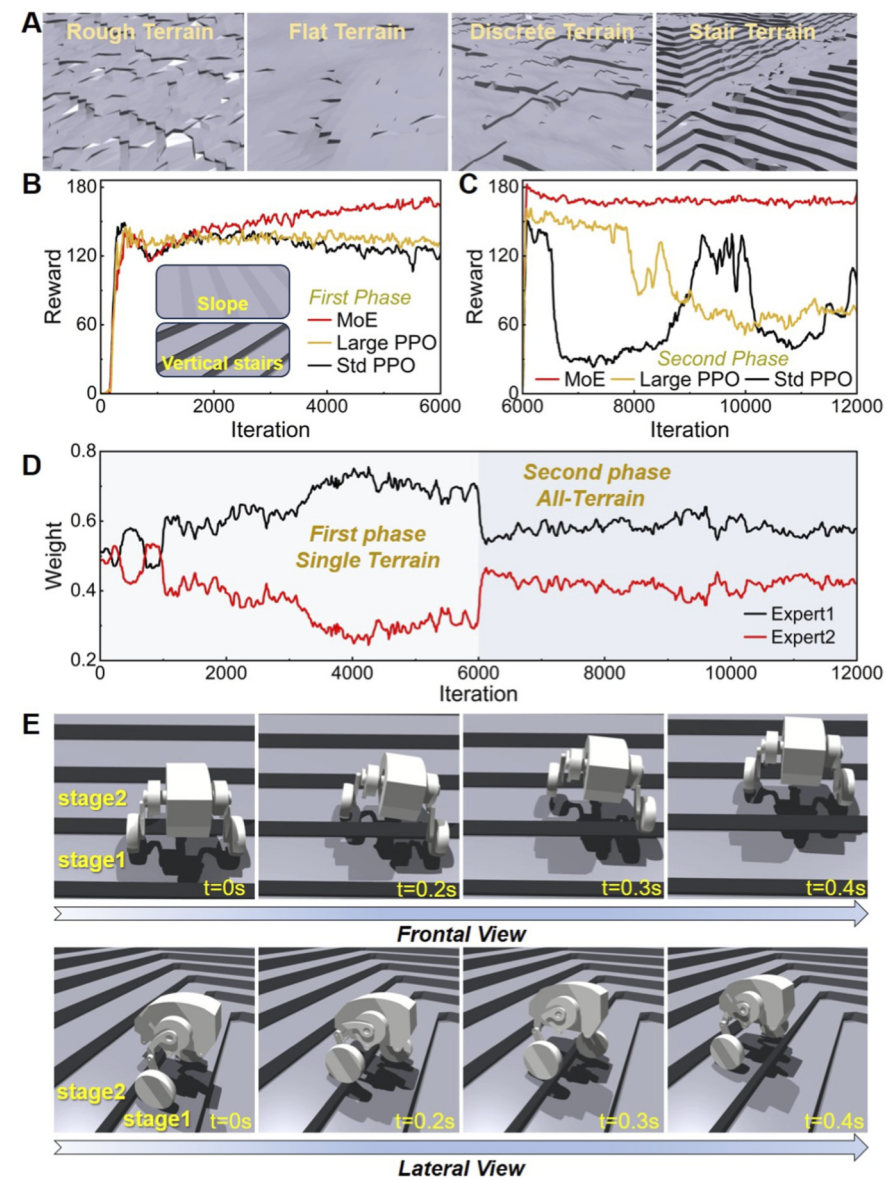


FIGURE 3
Terrain environment and stair climbing tests. (A) The training terrain after the addition of Perlin noise. (B) Comparison of training reward curves among the standard PPO baseline, large PPO baseline, and MoE in the first training phase. (C) Comparison of training reward curves among the standard PPO baseline, large PPO baseline, and MoE in the second training phase. (D) Average weight changes of the two expert networks throughout the training process. (E) Results of the robot using the trained locomotion policy to ascend stairs, with action decompositions shown from both frontal and lateral views.

validate the inherent limitations of single-network structures when handling multi-mode locomotion tasks in complex environments. The performance improvement of the MoE-enhanced algorithm arises primarily from its intrinsic mechanism, rather than the increase in total network parameters.

3.2 Experts contributions during training

The variation in the average weights of the experts during the training process of the MoE-enhanced algorithm was extracted in Figure 3D. It is evident that in the first phase, there was a substantial difference in the call frequencies between the two expert

networks; however, this disparity significantly decreased in the second phase. The primary reason for this change can be attributed to the relatively simple terrain in the first phase, which consisted solely of flat surfaces, slopes, and vertical stairs without any added Perlin noise. In this scenario, the use of the leg-lifting gait was required only for navigating the vertical stairs. When encountering vertical stairs, the robot needed to roll a certain distance on the step before transitioning to the next one, which meant that calls to the expert responsible for the leg-lifting gait were infrequent, occurring only during the transition from one stair to the next, while the majority of the time involved pure rolling. Consequently, during the first phase of training, the call frequency for the leg-lifting expert (Expert 2 in the figure) was significantly lower

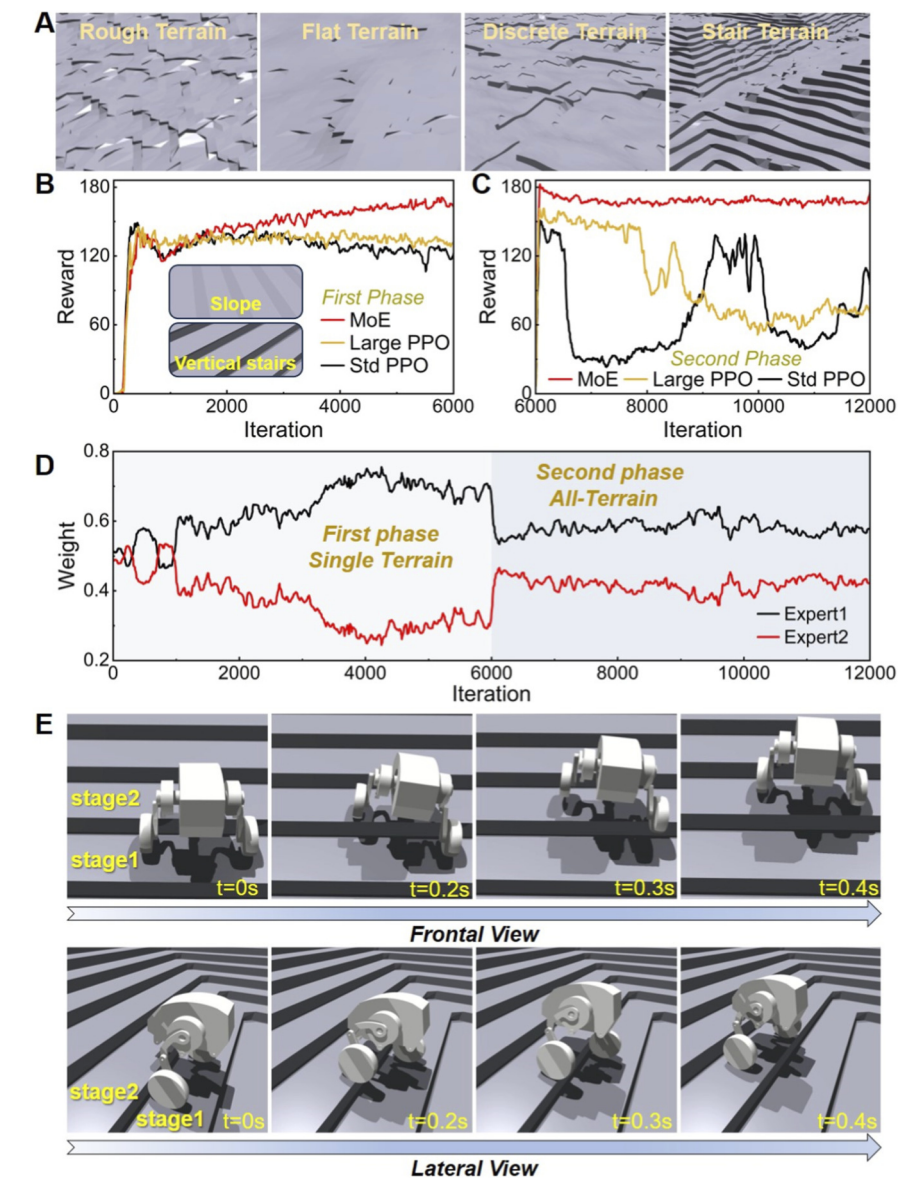


图3地形环境与爬楼梯测试。(A) 添加Perlin噪声后的训练地形。(B) 第一阶段训练中标准PPO基线、大型PPO基线与MoE的训练奖励曲线对比。(C) 第二阶段训练中标准PPO基线、大型PPO基线与MoE的训练奖励曲线对比。(D) 整个训练过程中两个专家网络的平均权重变化。(E) 机器人使用训练后的运动策略爬楼梯的结果，动作分解从正面和侧面视图展示。

验证了单网络结构在处理复杂环境中的多模式运动任务时固有的局限性。MoE增强算法的性能提升主要源于其内在机制，而非总网络参数的增加。

3.2 训练阶段的专家贡献

在MoE增强算法的训练过程中，提取了专家平均权重的变化，见图3D。可以看出，在第一阶段，两个专家之间的调用频率存在显著差异

网络；然而，这种差异在第二阶段显著减小。这一变化的主要原因可归因于第一阶段相对简单的地形，该阶段仅包含平坦表面、斜坡和垂直楼梯，未添加任何Perlin噪声。在这种情况下，仅在攀爬垂直楼梯时才需要使用抬腿步态。当遇到垂直楼梯时，机器人需要在台阶上滚动一定距离后才能过渡到下一级，这意味着负责抬腿步态的专家的调用并不频繁，仅在从一级楼梯过渡到下一级时才会发生，而大部分时间则处于纯滚动状态。因此，在训练的第一阶段，抬腿步态专家（图中专家2）的调用频率显著低于

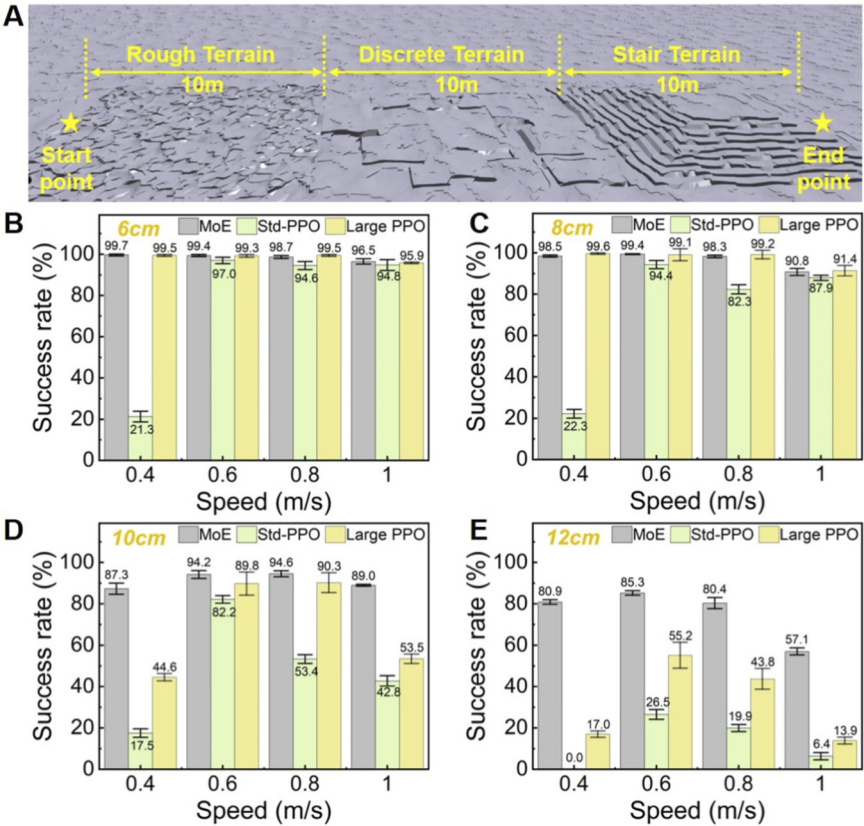


FIGURE 4 Obstacle navigation test. Success is defined as the robot traveling from the “Start point” to the “End point” without colliding with the ground and without straying outside the 10-m-wide test area. For each test statistic, 500 robots are tested independently, and the final success rate of these 500 robots is calculated. (A) Setup of the obstacle testing area. (B) Comparison of success rates for the three algorithms at a maximum vertical height of 6 cm. (C) Comparison of success rates for the three algorithms at a maximum vertical height of 8 cm. (D) Comparison of success rates for the three algorithms at a maximum vertical height of 10 cm. (E) Comparison of success rates for the three algorithms at a maximum vertical height of 12 cm.

than that of the expert responsible for pure rolling (Expert 1), leading to a notable difference in their average weights as the number of training epochs increased. In contrast, the second phase introduced discrete terrains that required frequent alternation between pure rolling and the leg-lifting gait, along with the addition of Perlin noise, which exacerbated the irregularity of the terrain surface. This heightened the demand for the leg-lifting gait. Under these terrain conditions, the call frequency for Expert 2 in the second phase noticeably increased to accommodate the undulating ground. As the number of training epochs in the second phase increased, the difference in average weights between the two experts significantly diminished.

3.3 Stair-climbing test

The locomotion policy derived from the MoE-enhanced algorithm was tested for ascending stairs, with the results presented in Figure 3E. When faced with stairs of a vertical height of 12 cm, the robot successfully executed an ideal leg-lifting gait. The motion breakdown was analyzed from both the frontal and lateral views, as shown in Figure 3E. With a speed command set at 0.6 m/s, the robot took approximately 0.4 s to clear the step. During the ascent, the robot first lifted its left wheel onto the next step before bringing up the right wheel, thereby effectively utilizing the

leg-lifting gait to complete the stair-climbing task. In general, for bipedal robots encountering vertical obstacles, common motion policies include jumping or employing a leg-lifting gait. However, in this study, due to the incorporation of a significant leg-lifting reward, the resulting obstacle negotiation policy was exclusively the leg-lifting gait. Utilizing only the leg-lifting gait for obstacle negotiation effectively addresses several issues associated with jumping, such as problems related to body stability and the impacts and vibrations that may affect the robot’s internal components.

3.4 Obstacle navigation test

A specially designed testing environment is established to test the obstacle negotiation success rates of the two policies (Figure 4A), which included three types of terrain: rough terrain, discrete terrain, and vertical stairs. Each terrain type was set within a square area, with the rough terrain generated using Perlin noise at an amplitude of 6 cm, and the maximum vertical heights for the discrete terrain and vertical stairs set between 6 and 12 cm. During the testing process, a successful trial required the robot to remain upright throughout, not exit the terrain boundaries, and reach the endpoint. Each terrain with distinct difficulty was measured repeatedly, and 500 robots were used in each single test for statistical

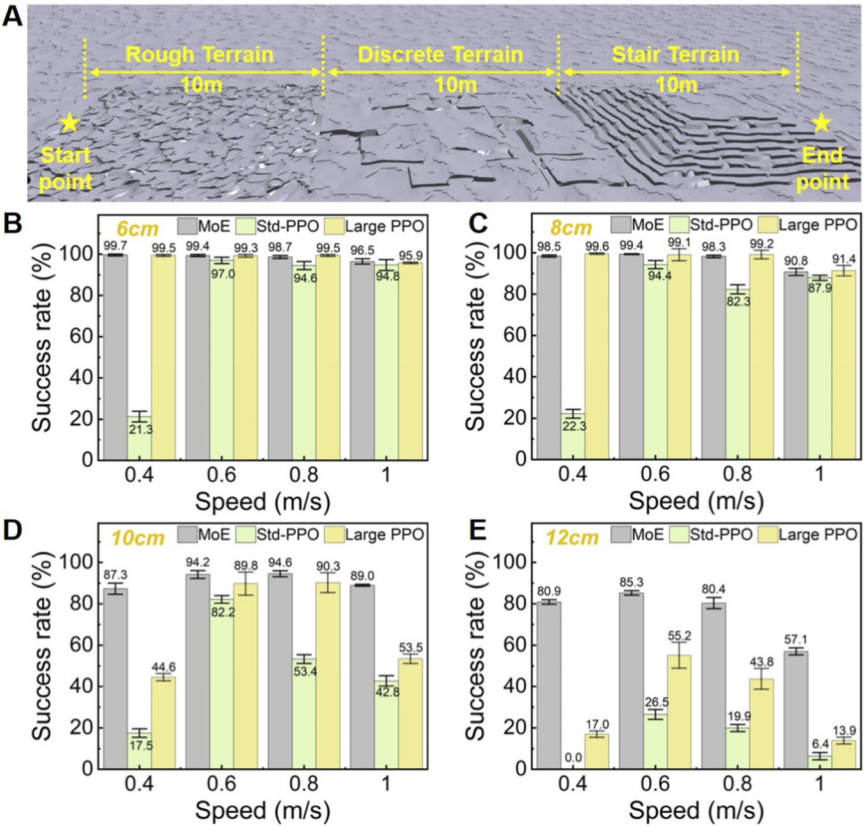


图4障碍物导航测试。成功的定义是机器人从“起点”行驶到“终点”，且不与地面碰撞、不偏离10米宽的测试区域。每项测试统计中，500台机器人独立测试，并计算这500台机器人的最终成功率。(A)障碍物测试区域设置。(B)最大垂直高度为6cm时三种算法的成功率比较。(C)最大垂直高度为8cm时三种算法的成功率比较。(D)最大垂直高度为10cm时三种算法的成功率比较。(E)最大垂直高度为12cm时三种算法的成功率比较。

负责纯滚动的专家（专家1），导致随着训练轮次的增加，两者的平均权重出现显著差异。相比之下，第二阶段引入了需要频繁在纯滚动和抬腿步态之间切换的离散地形，同时添加了Perlin噪声，加剧了地形表面的不规则性。这提高了对抬腿步态的需求。在这种地形条件下，第二阶段中专家2的调用频率明显增加，以适应起伏的地面。随着第二阶段训练轮次的增加，两位专家之间的平均权重差异显著缩小。

3.3 爬楼梯测试

基于MoE增强算法得出的运动策略被用于上楼梯测试，结果如图3E所示。面对垂直高度为12厘米的楼梯时，机器人成功执行了理想的抬腿步态。从正面和侧面视图对运动过程进行了分析，如图3E所示。在速度指令设定为0.6米/秒的情况下，机器人约用时0.4秒越过台阶。在上楼过程中，机器人先将左轮抬至下一级台阶，再抬起右轮，从而有效利用

抬腿步态完成爬楼梯任务。一般而言，对于遇到垂直障碍物的双足机器人，常见的运动策略包括跳跃或采用抬腿步态。然而，在本研究中，由于加入了显著的抬腿奖励，最终形成的障碍物协商策略仅为抬腿步态。仅采用抬腿步态进行障碍物协商，可以有效解决跳跃带来的若干问题，例如与身体稳定性相关的难题，以及可能影响机器人内部组件的冲击和振动。

3.4 障碍物导航测试

为测试两种策略的越障成功率，我们建立了一个专门设计的测试环境（图4A），其中包含三种地形：崎岖地形、离散地形和垂直楼梯。每种地形均设置在方形区域内，其中崎岖地形使用振幅为6厘米的Perlin噪声生成，离散地形和垂直楼梯的最大垂直高度设置在6至12厘米之间。在测试过程中，一次成功的试验要求机器人全程保持直立、不超出地形边界，并到达终点。每种不同难度的地形均进行了重复测量，每次单项测试使用500个机器人进行统计

analysis. The results are shown in [Figures 4B,C](#). From the results, the locomotion policy trained by the MoE-enhanced algorithm significantly outperforms the standard PPO baseline and the large PPO baseline. In terrains of all difficulty levels, its success rate exceeds 50% across all velocity commands. For the standard PPO baseline, insufficient obstacle-crossing performance can be clearly observed, especially under low velocity commands. Even in the simplest terrain with an obstacle height of 6 cm, its success rate remains low. In the terrain with a height of 12 cm, it performs poorly across all velocity commands. The large PPO baseline also exhibits a considerable performance gap compared with the MoE-enhanced algorithm, and this gap becomes increasingly pronounced as terrain difficulty rises. In particular, when the maximum vertical obstacle height reaches 12 cm, the success rate of the large PPO baseline is generally below 50%. Overall, these results further demonstrate that the superiority of the MoE-enhanced algorithm mainly stems from the inherent mechanism of MoE, and is not significantly affected by the specific amount of network parameters.

Based on this experiment, we further conducted ablation experiments to analyze the effects of MoE and phased training. We established an additional baseline trained directly for 12,000 iterations without phasing. Similarly, we evaluate the two training schemes using two metrics: the average training reward and the obstacle-crossing success rate on terrains of different heights. First, with respect to the average reward, training was performed with four different random seeds, and the reward results were statistically analyzed, as shown in [Supplementary Figure S2](#) in the [Supplementary Material](#). It can be clearly observed that the MoE-enhanced algorithm achieves higher reward values under phased training than under non-phased training. Subsequently, the locomotion policies obtained by the two training methods were tested for obstacle-crossing success rate, with the results presented in [Supplementary Figure S3](#) in the [Supplementary Material](#). The results show that the performance difference between the two policies is not significant on relatively simple terrains. When the maximum vertical height of obstacles reaches 12 cm, the success rate of the policy trained with phased training is significantly higher than that of the non-phased policy. This validates that phased training can improve the upper bound of obstacle-crossing performance. Combined with the comparison results against the PPO algorithm using a single network, it can be concluded that in the MoE-enhanced algorithm proposed in this study, the mechanism of MoE itself plays a more dominant role, while phased training serves as an auxiliary approach to further improve the algorithm's performance.

4 Conclusion

This study proposes a reinforcement learning control framework that integrates the Mixture of Experts (MoE) for bipedal wheel-legged robots. By incorporating MoE into the Actor network of the PPO algorithm, we leverage a sparse gating mechanism to decouple parameters for pure rolling and leg-lifting tasks, effectively resolving the gradient conflict issues associated with single-network multi-task learning. Simulation test results demonstrate that this approach not only eliminates catastrophic forgetting but also significantly enhances training stability. To ensure the robustness and generalization capabilities of the policy, we implemented a two-phase

curriculum learning strategy that transitions from specific terrains to complex terrains with Perlin noise. The MoE-enhanced strategy exhibits remarkable adaptability when navigating complex terrains, automatically adjusting expert weights according to the terrain's undulations, thus achieving a smooth transition from rolling on flat ground to dynamic obstacle negotiation. Notably, in high-difficulty obstacle tests, this policy achieved a success rate that greatly exceeded that of the single-network PPO algorithm when traversing vertical stairs of 12 cm. Future work will focus on several key areas: recognizing the significant impact of collision forces on balance during high-speed movements, further research will aim to enhance the system's active safety and self-recovery capabilities under extreme physical contact and external disturbances. Additionally, we will conduct real-world deployment tests in more diverse and unstructured outdoor environments to validate and optimize the long-term reliability of this method in practical applications.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Author contributions

PH: Writing – review and editing, Writing – original draft. ZZ: Writing – review and editing, Writing – original draft. SD: Methodology, Conceptualization, Writing – review and editing. PW: Writing – review and editing, Investigation. HL: Project administration, Methodology, Writing – review and editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

分析。结果如图4B、C所示。从结果来看，由MoE增强算法训练的运动策略显著优于标准PPO基线和大型PPO基线。在所有难度等级的地形中，该策略在所有速度指令下的成功率均超过50%。对于标准PPO基线，可以明显观察到其越障性能不足，尤其是在低速度指令下。即使在障碍物高度为6厘米的最简单地形中，其成功率仍然很低。在高度为12厘米的地形中，该策略在所有速度指令下均表现不佳。大型PPO基线相比MoE增强算法也存在相当大的性能差距，且随着地形难度的增加，这一差距愈发明显。特别是当最大垂直障碍物高度达到12厘米时，大型PPO基线的成功率普遍低于50%。总体而言，这些结果进一步证明，MoE增强算法的优越性主要源于MoE的固有机制，而不受网络参数具体数量的显著影响。

基于该实验，我们进一步开展了消融实验，以分析MoE和分阶段训练的效果。我们建立了一个额外的基线，该基线直接训练12,000次迭代而不进行分阶段。同样，我们使用两个指标来评估这两种训练方案：平均训练奖励和在不同高度地形上的越障成功率。首先，关于平均奖励，我们使用四种不同的随机种子进行训练，并对奖励结果进行了统计分析，如[补充图S2](#)中[补充材料](#)所示。可以清楚地观察到，MoE增强算法在分阶段训练下比非分阶段训练获得了更高的奖励值。随后，对两种训练方法获得的运动策略进行了越障成功率测试，结果见[补充图S3](#)中[补充材料](#)。结果表明，在相对简单的地形上，两种策略的性能差异不显著。当障碍物的最大垂直高度达到12厘米时，采用分阶段训练的策略的成功率显著高于非分阶段策略。这验证了分阶段训练可以提高越障性能的上界。结合与使用单一网络的PPO算法的对比结果，可以得出结论：在本研究提出的MoE增强算法中，MoE机制本身起着更主导的作用，而分阶段训练则是进一步提高算法性能的辅助手段。

4 结论

本研究提出了一种面向双足轮腿机器人的强化学习控制框架，该框架集成了专家混合（MoE）。通过在PPO算法的Actor网络中引入MoE，我们利用稀疏门控机制对纯滚动和抬腿任务的参数进行解耦，有效解决了单网络多任务学习中的梯度冲突问题。仿真测试结果表明，该方法不仅消除了灾难性遗忘，还显著提升了训练稳定性。为确保策略的鲁棒性和泛化能力，我们实施了一个两阶段

课程学习策略，该策略从特定地形逐步过渡到带有Perlin噪声的复杂地形。MoE增强策略在应对复杂地形时展现出卓越的适应性，能够根据地形的起伏自动调整专家权重，从而实现从平地滚动到动态障碍物协商的平滑过渡。值得注意的是，在高难度障碍物测试中，该策略在穿越12厘米垂直楼梯时的成功率远超单网络PPO算法。未来的工作将聚焦于以下几个关键方向：认识到高速运动过程中碰撞力对平衡的显著影响后，进一步研究将致力于增强系统在极端物理接触和外部扰动下的主动安全与自恢复能力。此外，我们将在更多样化、更非结构化的户外环境中进行实际部署测试，以验证并优化该方法在实际应用中的长期可靠性。

数据可用性声明

支持本文结论的原始数据将由作者提供，无需过多保留。

作者贡献

PH: 写作——审阅与编辑，写作——初稿。ZZ: 写作——审阅与编辑，写作——初稿。SD: 方法论，概念化，写作——审阅与编辑。PW: 写作——审阅与编辑，调查。HL: 项目管理，方法论，写作——审阅与编辑。

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